

Top 10 Mobile Penetration Testing Tools for Ethical Hackers

Build a mobile pentest workflow that's fast, practical, and repeatable.



By HexSec

Feature highlights:

- Rapid static + dynamic visibility
- Inspect real app logic, not assumptions
- Validate findings with traffic and runtime

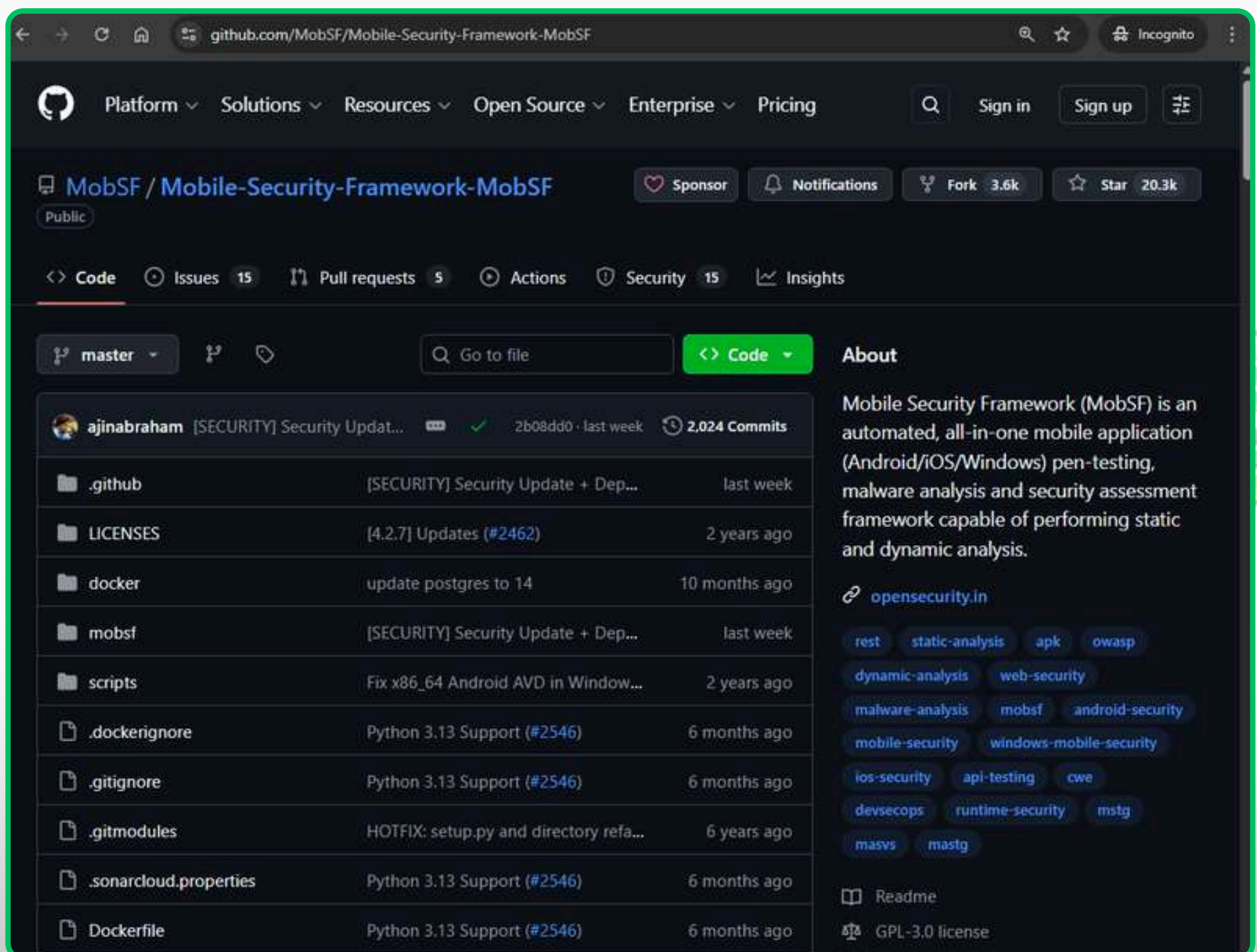
If you only master 10 mobile tools this year, make them these.



MobSF

1

MobSF is an automated, all-in-one framework for assessing mobile apps. It combines static analysis (APKs/IPAs), dynamic testing, and basic malware checks into one dashboard, generating clear findings and exportable reports. It's ideal for fast triage and for building a repeatable lab workflow you can run locally, in Docker, or inside CI when reviewing many apps under time pressure. Plus evidence-ready outputs.



 @hex_sec
  hexsec_tools
  hexsecteam

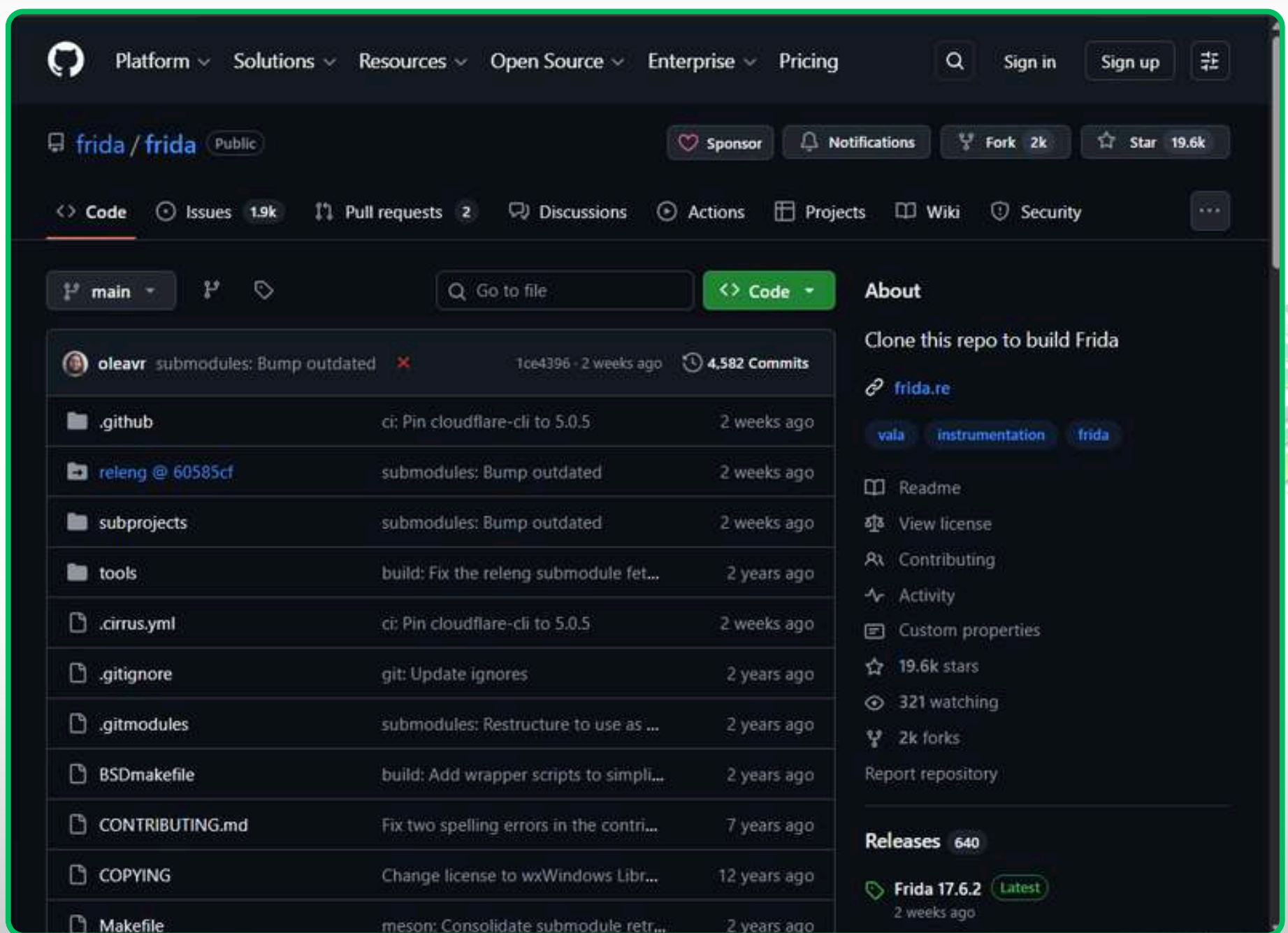
<https://github.com/HackTricks-wiki/hacktricks>



Frida

2

Frida is a dynamic instrumentation toolkit that lets you hook and inspect app behavior at runtime. You can trace functions, alter return values, and observe sensitive operations while the app runs—perfect for validating security controls and understanding hidden flows. It's widely used because it works across mobile and desktop targets and scales from quick probes to full custom scripts.



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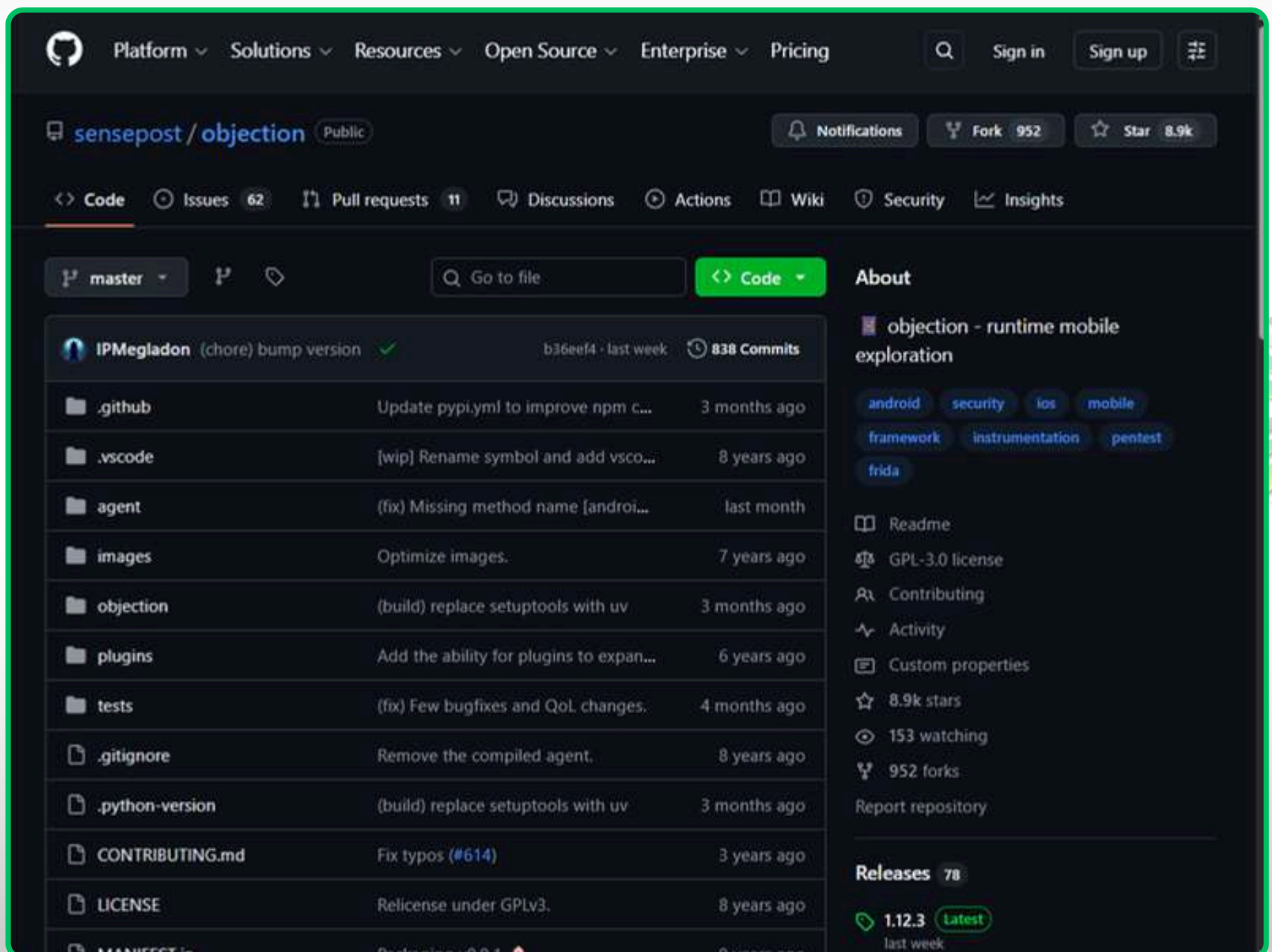
<https://github.com/frida/frida>



Objection

3

Objection is a Frida-powered mobile exploration toolkit that wraps common runtime tasks into easy commands. It helps you enumerate app internals, inspect storage, explore classes, and interact with running processes without writing boilerplate. When you need momentum in a mobile assessment, it turns “setup pain” into quick wins and helps you iterate on hypotheses in minutes. With minimal friction.



The screenshot shows the GitHub repository page for `sensepost/objection`. The repository is public and has 952 forks and 8.9k stars. The main content area displays a list of files and folders, including `.github`, `.vscode`, `agent`, `images`, `objection`, `plugins`, `tests`, `.gitignore`, `.python-version`, `CONTRIBUTING.md`, `LICENSE`, and `MANIFEST.in`. The right sidebar contains the "About" section, which describes the project as "objection - runtime mobile exploration" and lists tags such as `android`, `security`, `ios`, `mobile`, `framework`, `instrumentation`, `pentest`, and `frida`. The "Releases" section shows the latest release, `1.12.3`, which was published last week.

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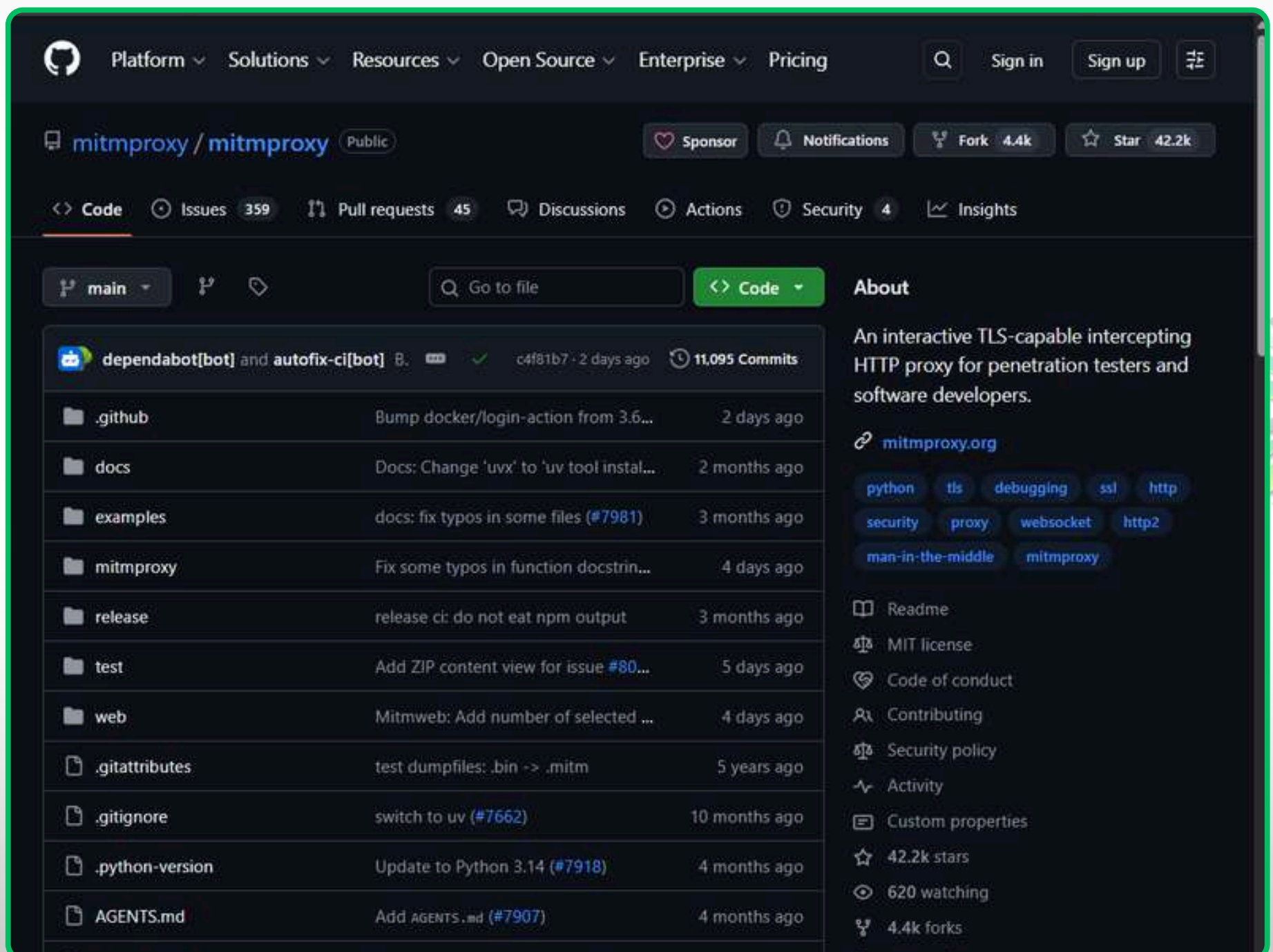
<https://github.com/sensepost/objection>



Mitmproxy

4

mitmproxy is a strong open-source intercepting proxy for HTTP(S) traffic. It lets you capture, inspect, modify, and replay requests/responses, making it easier to verify authentication, session handling, API behavior, and data exposure. With scripting and addons, you can automate repeatable checks, craft edge-case payloads, and keep clean evidence of what the app actually sends over the wire.



The screenshot shows the GitHub repository for mitmproxy. The repository is public and has 42.2k stars, 4.4k forks, and 620 watchers. It is maintained by dependabot[bot] and autofix-ci[bot]. The repository contains a README, MIT license, Code of conduct, Contributing guide, Security policy, and Activity. The repository is described as an interactive TLS-capable intercepting HTTP proxy for penetration testers and software developers. The repository is available at <https://github.com/mitmproxy/mitmproxy>.

File/Folder	Commit Message	Commit Date
.github	Bump docker/login-action from 3.6...	2 days ago
docs	Docs: Change 'uvx' to 'uv tool instal...	2 months ago
examples	docs: fix typos in some files (#7981)	3 months ago
mitmproxy	Fix some typos in function docstrin...	4 days ago
release	release ci: do not eat npm output	3 months ago
test	Add ZIP content view for issue #80...	5 days ago
web	Mitmweb: Add number of selected ...	4 days ago
.gitattributes	test dumpfiles: .bin -> .mitm	5 years ago
.gitignore	switch to uv (#7662)	10 months ago
.python-version	Update to Python 3.14 (#7918)	4 months ago
AGENTS.md	Add AGENTS.md (#7907)	4 months ago

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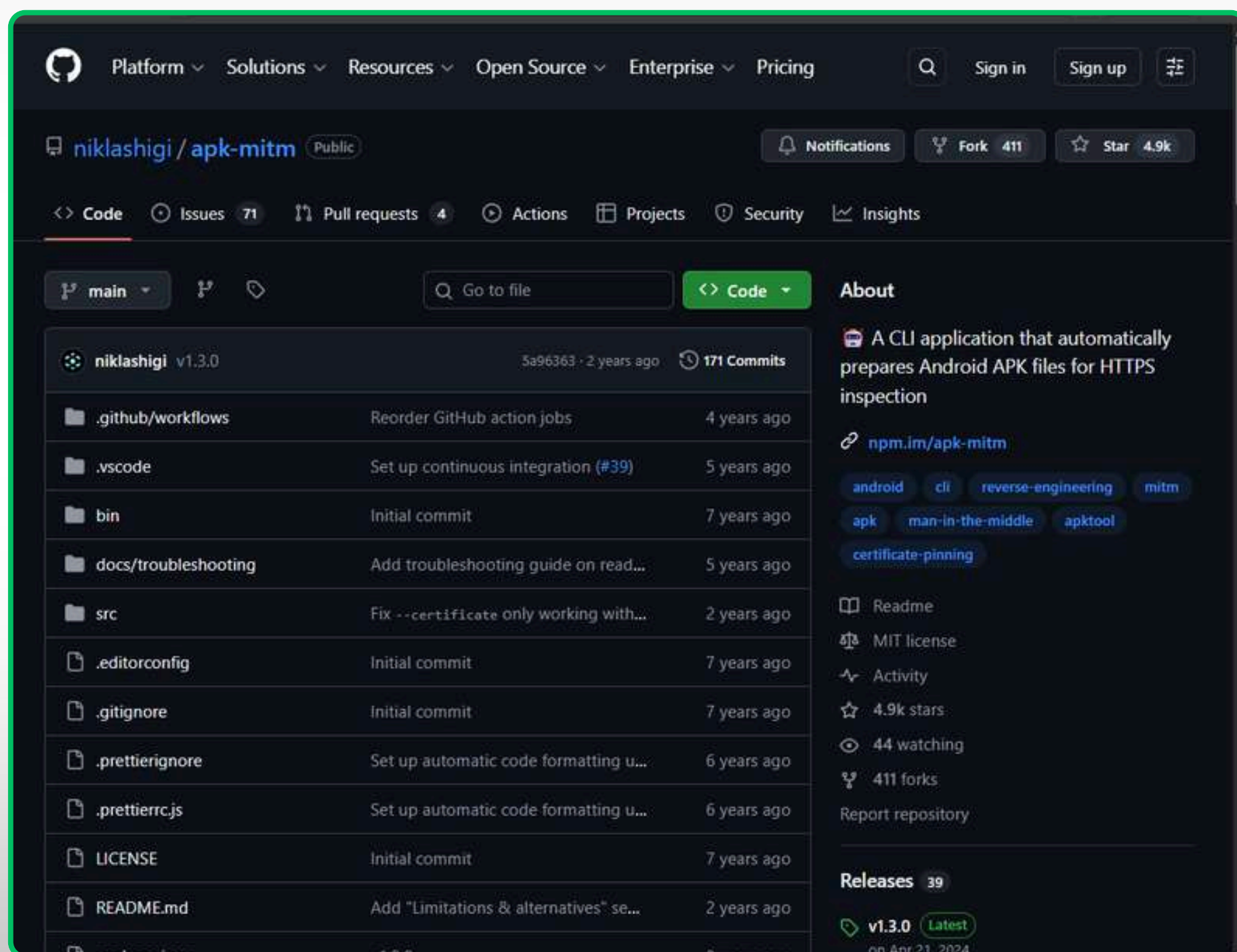
<https://github.com/mitmproxy/mitmproxy>



apk-mitm

5

apk-mitm is a CLI that prepares Android APKs for HTTPS inspection by automating the tedious parts of proxy setup. It patches apps to trust your interception certificate and smooths common blockers introduced by modern Android network security settings, so you can focus on analysis instead of fighting tooling. For many real-world apps, it's the quickest route from "can't intercept" to "traffic visible".



The screenshot shows the GitHub repository page for `niklashigi/apk-mitm`. The repository is public and has 411 forks and 4.9k stars. The main branch is selected. The repository contains a list of files and folders, including `.github/workflows`, `.vscode`, `bin`, `docs/troubleshooting`, `src`, `.editorconfig`, `.gitignore`, `.prettierrc`, `.prettierrc.js`, `LICENSE`, and `README.md`. The repository is described as a CLI application that automatically prepares Android APK files for HTTPS inspection. The latest release is v1.3.0, published on Apr 21, 2024.

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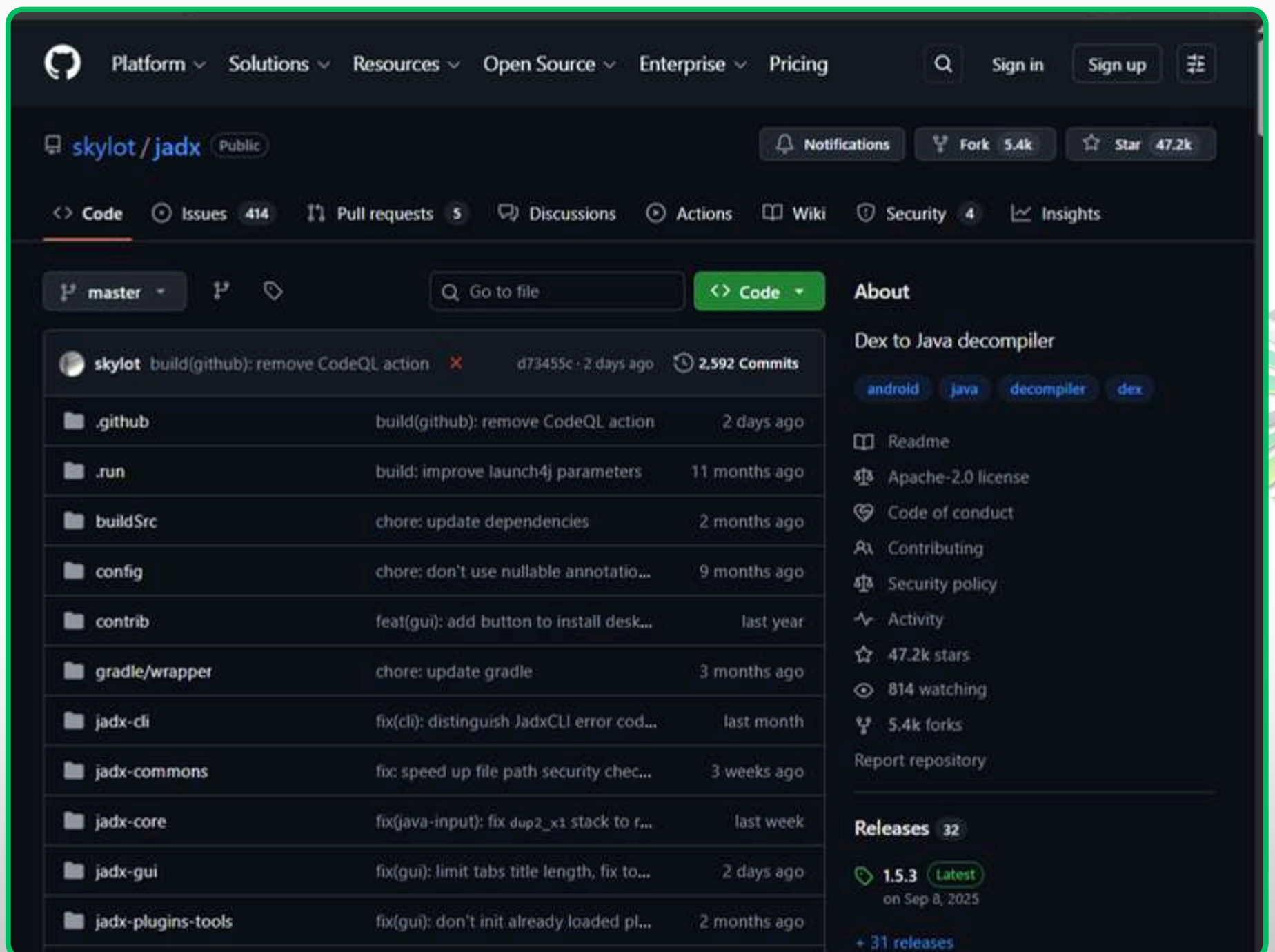
<https://github.com/niklashigi/apk-mitm>



JADX

6

JADX is a popular Dex-to-Java decompiler with both GUI and CLI. It helps you read Android app logic, spot insecure patterns, and navigate code paths quickly with search and cross-references. It's one of the fastest ways to understand an APK's architecture, identify interesting classes, and map what to test dynamically and on the network—especially when source code is unavailable. For quicker review.



The screenshot shows the GitHub repository for `skylot/jadx`. The repository is public and has 47.2k stars, 5.4k forks, and 414 issues. The main branch is `master`. The repository contains several folders and files, including `.github`, `.run`, `buildSrc`, `config`, `contrib`, `gradle/wrapper`, `jadx-cli`, `jadx-commons`, `jadx-core`, `jadx-gui`, and `jadx-plugins-tools`. The repository is described as a "Dex to Java decompiler" and includes a README, Apache-2.0 license, Code of conduct, Contributing, Security policy, and Activity. The latest release is 1.5.3, released on Sep 8, 2025.

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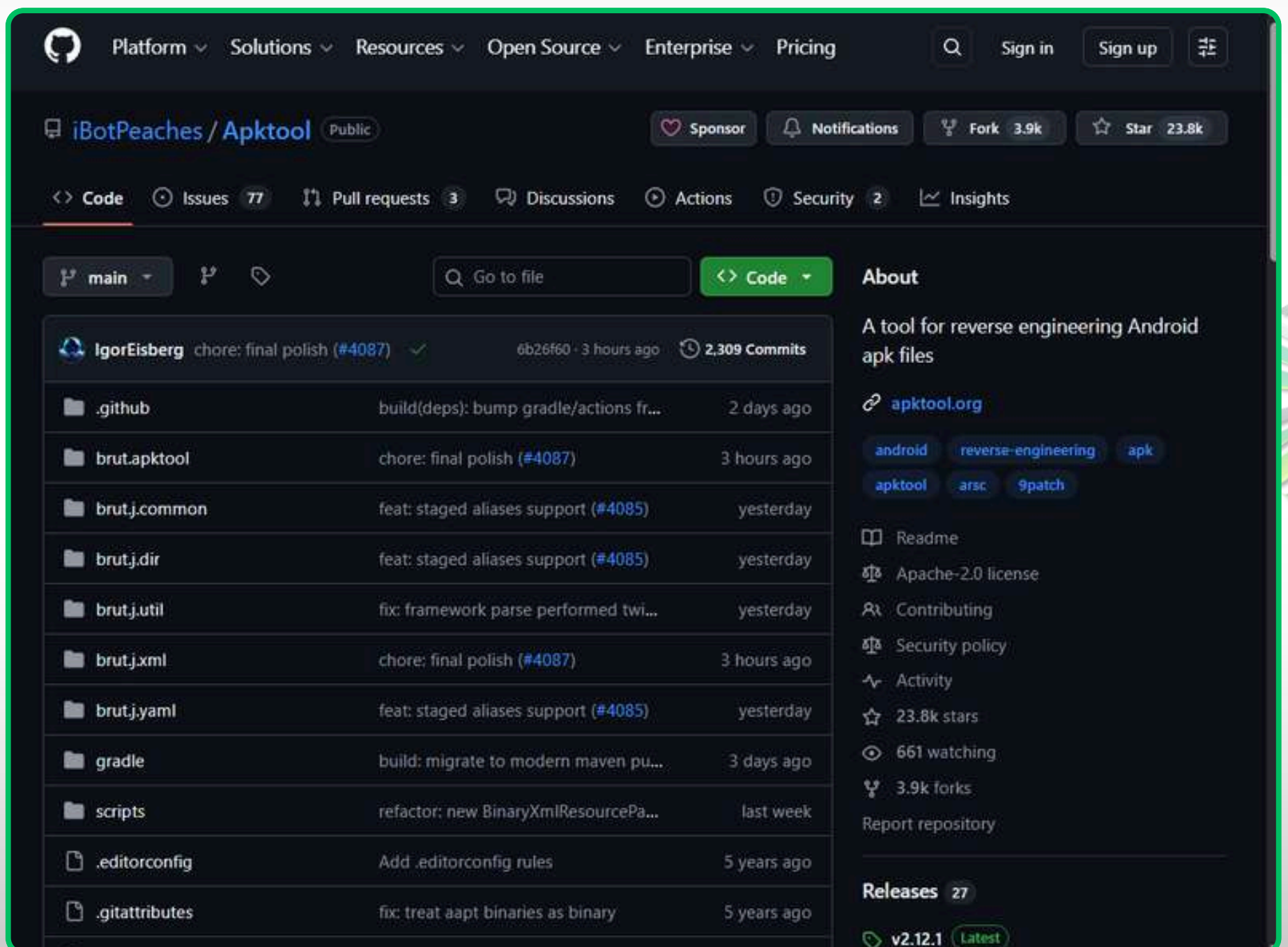
<https://github.com/skylot/jadx>



Apktool

7

Apktool reverse-engineers Android APK resources and smali, letting you decode, modify, and rebuild apps. It's essential for inspecting manifests, permissions, and resource configs, and for bytecode-level changes when you need a controlled test build. It also helps confirm what the packaged app truly declares and ships, which is key when debugging security behavior across versions. During regression tests.



The screenshot shows the GitHub repository for iBotPeaches/Apktool. The repository is public and has 23.8k stars, 3.9k forks, and 77 issues. The main branch is selected. The repository description is "A tool for reverse engineering Android apk files". The repository is licensed under Apache-2.0. The repository has 27 releases, with the latest release being v2.12.1. The repository is categorized under android, reverse-engineering, apk, apktool, arsc, and 9patch. The repository is also categorized under Readme, Apache-2.0 license, Contributing, Security policy, Activity, 23.8k stars, 661 watching, 3.9k forks, and Report repository.

File/Folder	Commit Message	Commit Time
.github	build(deps): bump gradle/actions fr...	2 days ago
brut.apktool	chore: final polish (#4087)	3 hours ago
brut.j.common	feat: staged aliases support (#4085)	yesterday
brut.j.dir	feat: staged aliases support (#4085)	yesterday
brut.j.util	fix: framework parse performed twi...	yesterday
brut.j.xml	chore: final polish (#4087)	3 hours ago
brut.j.yaml	feat: staged aliases support (#4085)	yesterday
gradle	build: migrate to modern maven pu...	3 days ago
scripts	refactor: new BinaryXmlResourcePa...	last week
.editorconfig	Add .editorconfig rules	5 years ago
.gitattributes	fix: treat aapt binaries as binary	5 years ago

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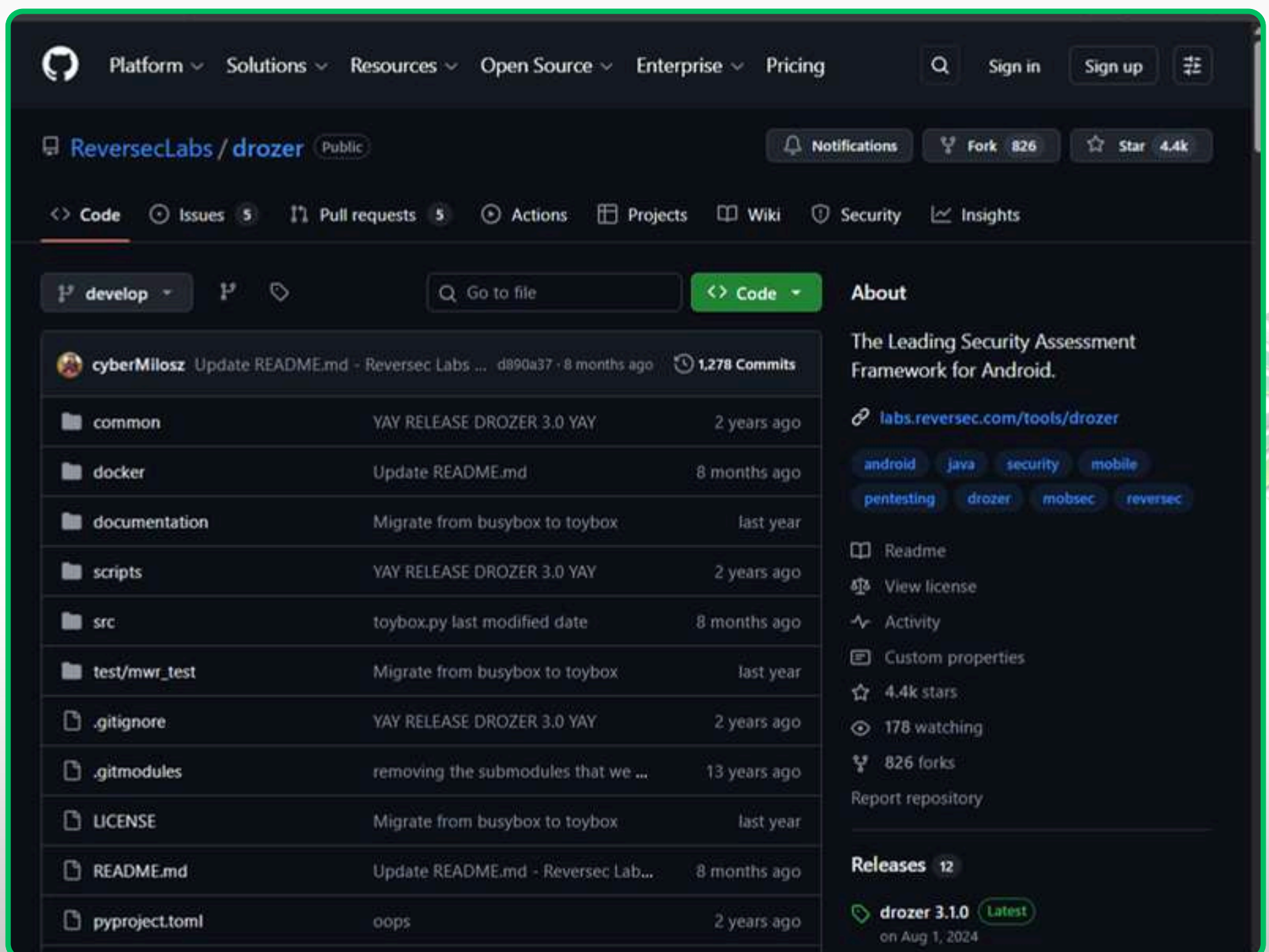
<https://github.com/iBotPeaches/Apktool>



Drozer

8

Drozer is an Android security assessment framework focused on IPC and component exposure. It lets you interact with exported activities, services, broadcast receivers, and content providers to identify misconfigurations and risky entry points. It's especially useful for testing app-to-app boundaries and OS interactions, surfacing issues like unintended data leaks or overly permissive components.



The screenshot shows the GitHub repository for **ReversesecLabs/drozer**. The repository is public and has 4.4k stars, 826 forks, and 178 watchers. The main branch is **develop**. The repository contains the following files and folders:

File/Folder	Last Commit	Time Ago
common	YAY RELEASE DROZER 3.0 YAY	2 years ago
docker	Update README.md	8 months ago
documentation	Migrate from busybox to toybox	last year
scripts	YAY RELEASE DROZER 3.0 YAY	2 years ago
src	toybox.py last modified date	8 months ago
test/mwr_test	Migrate from busybox to toybox	last year
.gitignore	YAY RELEASE DROZER 3.0 YAY	2 years ago
.gitmodules	removing the submodules that we ...	13 years ago
LICENSE	Migrate from busybox to toybox	last year
README.md	Update README.md - Reversesec Lab...	8 months ago
pyproject.toml	oops	2 years ago

The repository is described as "The Leading Security Assessment Framework for Android." and includes tags for **android**, **java**, **security**, **mobile**, **pentesting**, **drozer**, **mobsec**, and **reversesec**. The latest release is **drozer 3.1.0**, marked as "Latest" and released on Aug 1, 2024.

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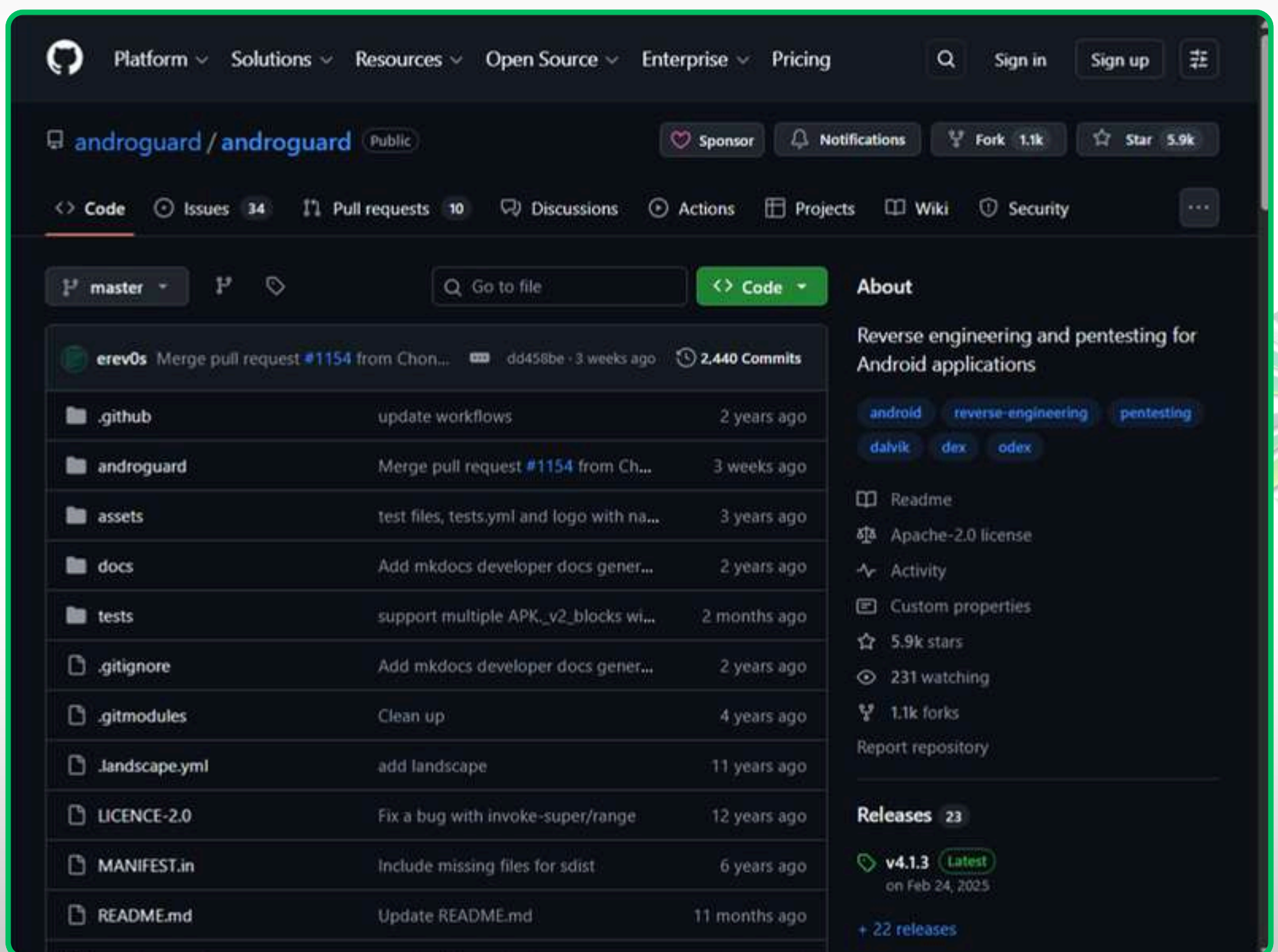
<https://github.com/ReversesecLabs/drozer>



Androguard

9

Androguard is a Python toolkit for reverse engineering and analyzing Android apps. It offers APIs to parse DEX, inspect bytecode, analyze manifests, and build custom scripts that extract insights at scale. It's great for automation: batch triage, custom rules, and enrichment of findings you discovered in JADX or MobSF, turning manual observations into repeatable checks across multiple APKs.



The screenshot shows the GitHub repository for Androguard. The repository is public and has 5.9k stars, 1.1k forks, and 34 issues. The main branch is master. The repository contains the following files and folders:

File/Folder	Description	Last Commit
.github	update workflows	2 years ago
androguard	Merge pull request #1154 from Chon...	3 weeks ago
assets	test files, tests.yml and logo with na...	3 years ago
docs	Add mkdocs developer docs gener...	2 years ago
tests	support multiple APK_v2_blocks wi...	2 months ago
.gitignore	Add mkdocs developer docs gener...	2 years ago
.gitmodules	Clean up	4 years ago
.landscape.yml	add landscape	11 years ago
LICENCE-2.0	Fix a bug with invoke-super/range	12 years ago
MANIFEST.in	Include missing files for sdist	6 years ago
README.md	Update README.md	11 months ago

The repository is about reverse engineering and pentesting for Android applications. It includes tags for android, reverse-engineering, pentesting, dalvik, dex, and odex. The repository has 5.9k stars, 231 watching, and 1.1k forks. The latest release is v4.1.3, released on Feb 24, 2025.

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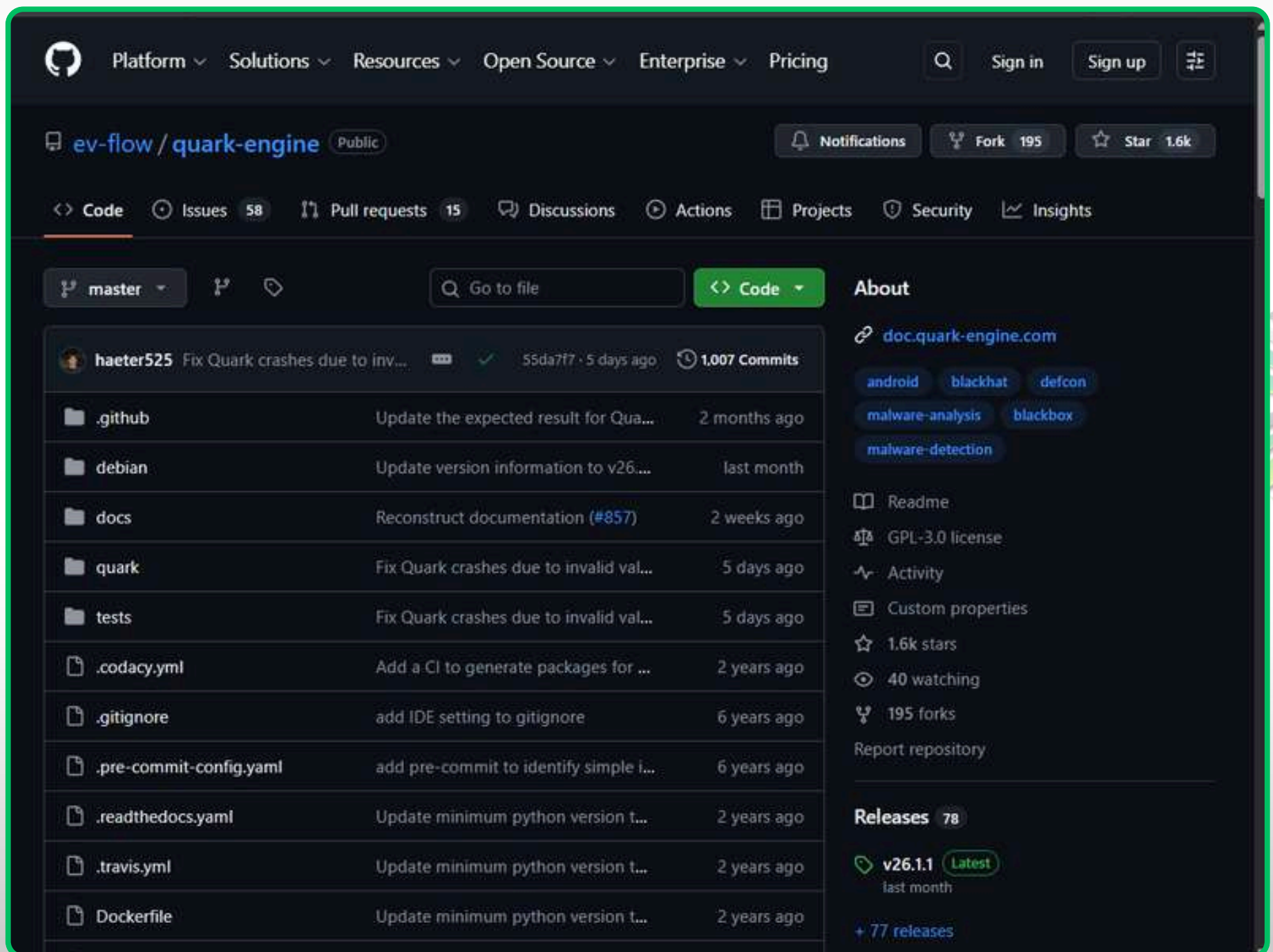
<https://github.com/androguard/androguard>



Quark Engine

10

Quark Engine is an open-source Android APK analysis engine that matches app behavior against a maintained rule set to highlight suspicious or risky patterns. It produces readable reports and helps you spot red flags faster, especially with large APKs or obfuscated code. Use it to prioritize what deserves deeper reversing, then validate the top signals with dynamic testing and traffic inspection.

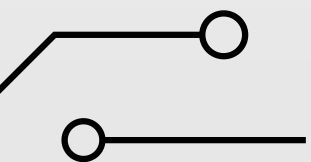


The screenshot shows the GitHub repository page for `ev-flow/quark-engine`. The repository is public and has 1.6k stars, 195 forks, and 40 watchers. The main branch is `master`. The repository contains several files and folders, including `.github`, `debian`, `docs`, `quark`, `tests`, `.codacy.yml`, `.gitignore`, `.pre-commit-config.yaml`, `.readthedocs.yaml`, `.travis.yml`, and `Dockerfile`. The repository is licensed under GPL-3.0. The latest release is `v26.1.1`, which was released last month. The repository is also available on `doc.quark-engine.com`.

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<https://github.com/ev-flow/quark-engine>





FINAL TIPS

Start with MobSF for a quick baseline, then validate the interesting findings by reading the app in JADX and Apktool. Treat static results as hypotheses: note risky flows, endpoints, and storage you must confirm.

Move to traffic testing with mitmproxy. If interception is blocked on Android, use apk-mitm to prepare a test build so you can see real request/response behavior. Save clean evidence and re-test after each small change.

Finish with runtime checks: use Frida + Objection to observe what the app actually does, and use Drozer to probe exposed Android components. Use Androguard and Quark Engine to automate triage and prioritize what deserves deeper reversing across large app sets.



@hex_sec



hexsec_tools



hexsecteam

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